Project Overview: Static Website (My Resume) Hosting on AWS

View on Web: https://hardikaws.online

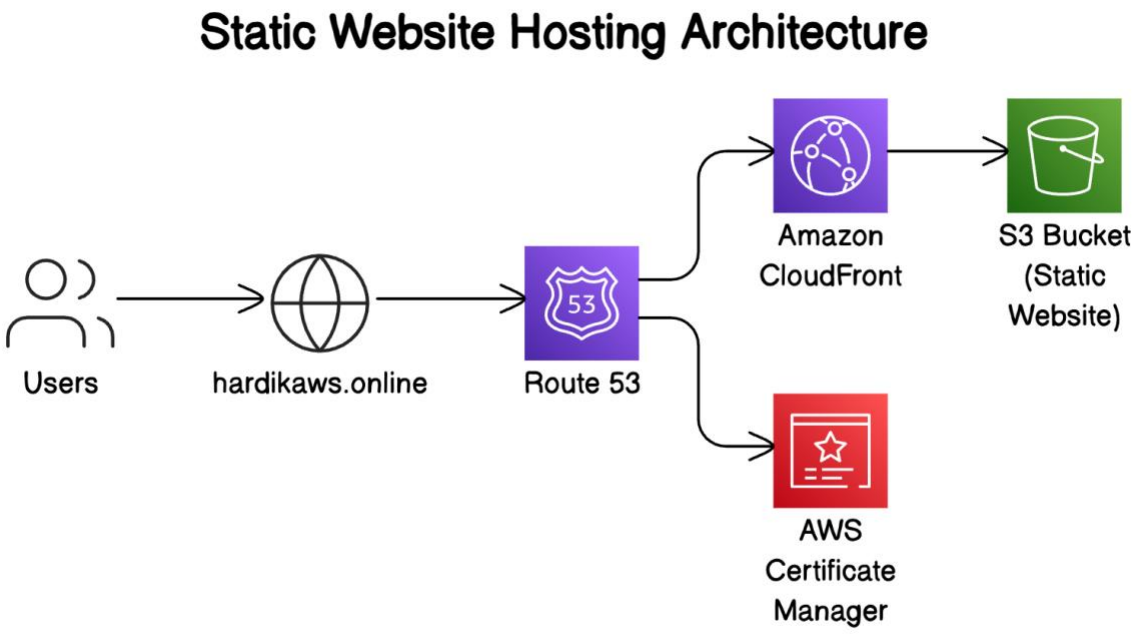
Objective:

To create a professional online presence by hosting a resume website using AWS services, demonstrating proficiency in cloud computing and web technologies.

Services Used:

AWS S3
AWS Route 53
AWS CloudFront
AWS ACM
GoDaddy

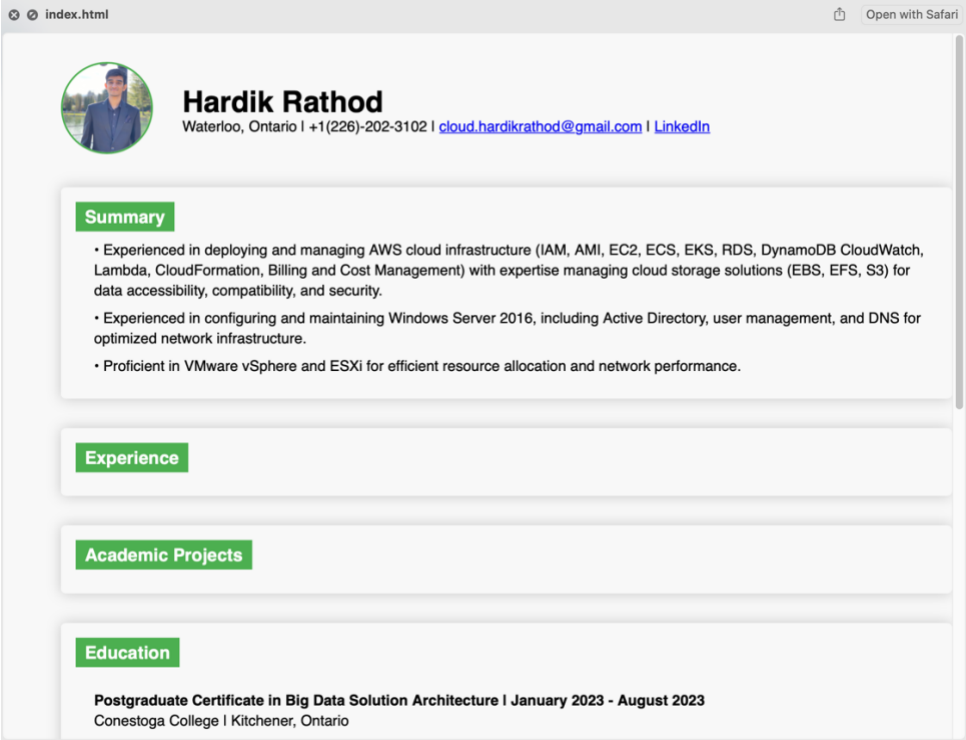
Architecture Diagram:



Implementation Steps:

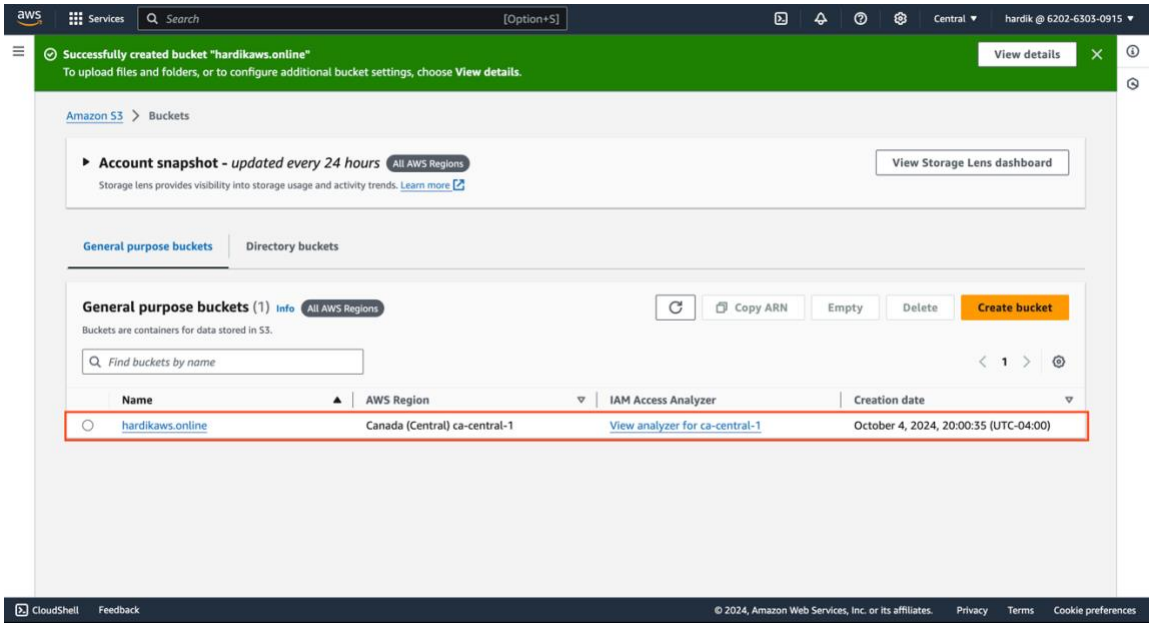
1. Web Development:

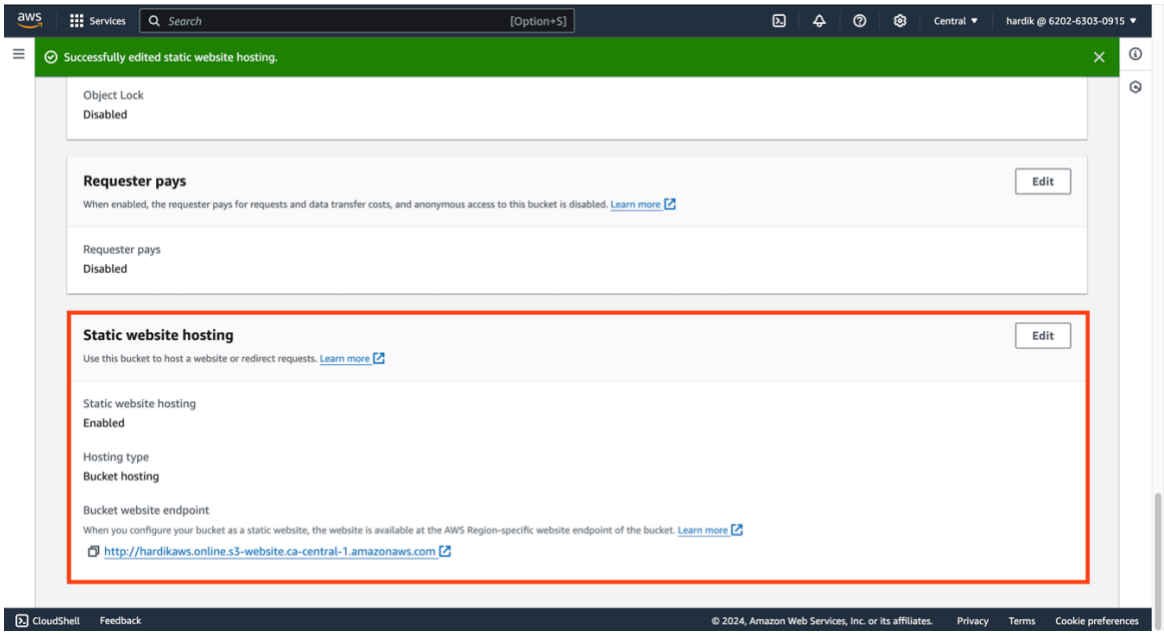
- Developed a responsive resume website using HTML, CSS, and JavaScript.
- Included a professional headshot image to enhance the visual appeal of the site.



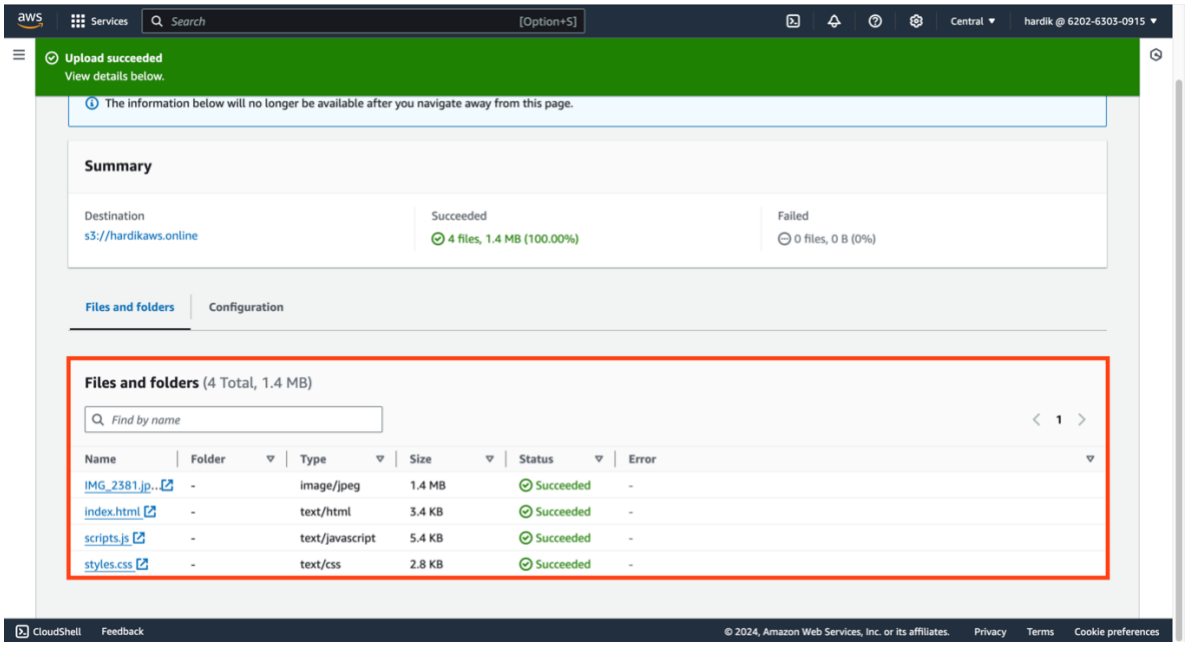
2. AWS S3 Configuration:

- Created an Amazon S3 bucket and configured it for static website hosting with public access.

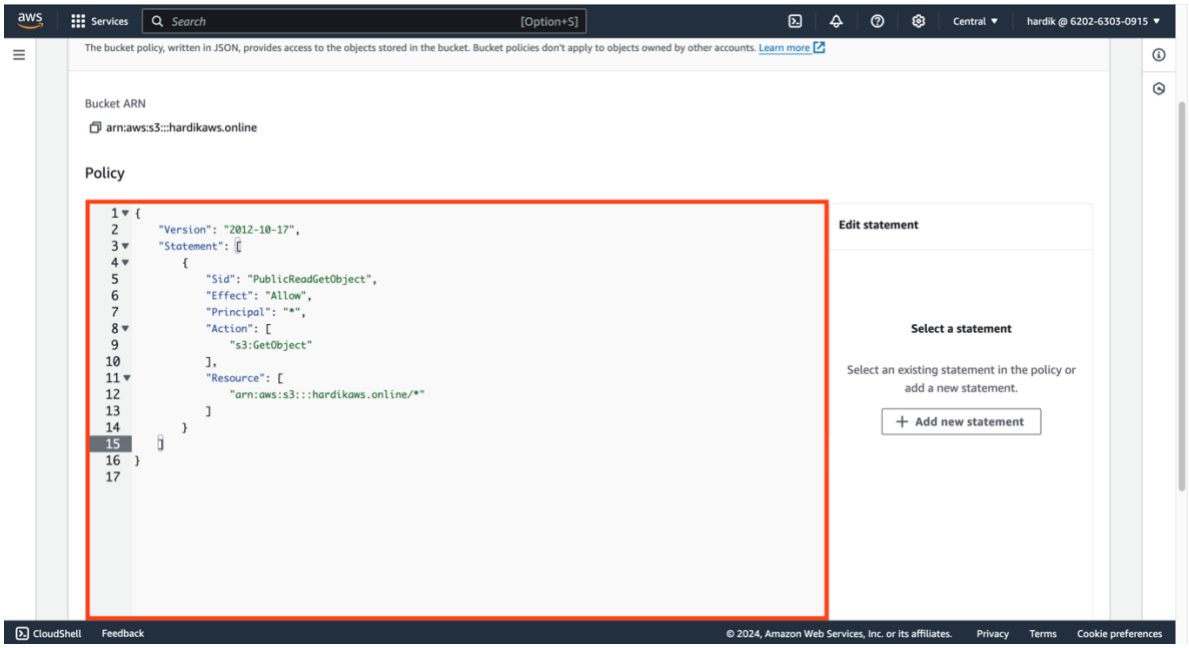




- Uploaded the HTML, CSS, JavaScript files, and the headshot image to the S3 bucket.

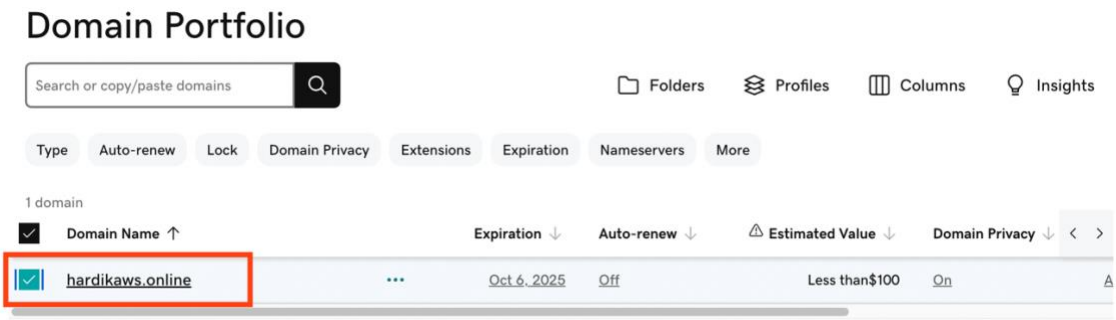


- Implemented a bucket policy to enable public read access to the uploaded files.

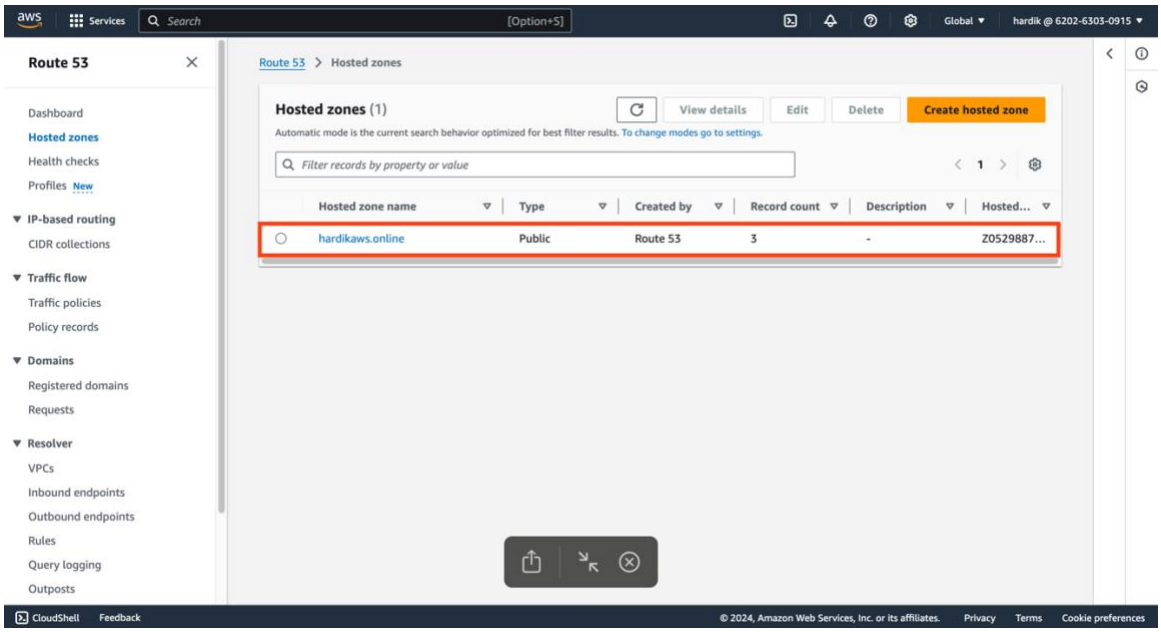


3. Domain Registration:

- Registered a custom domain name through GoDaddy.

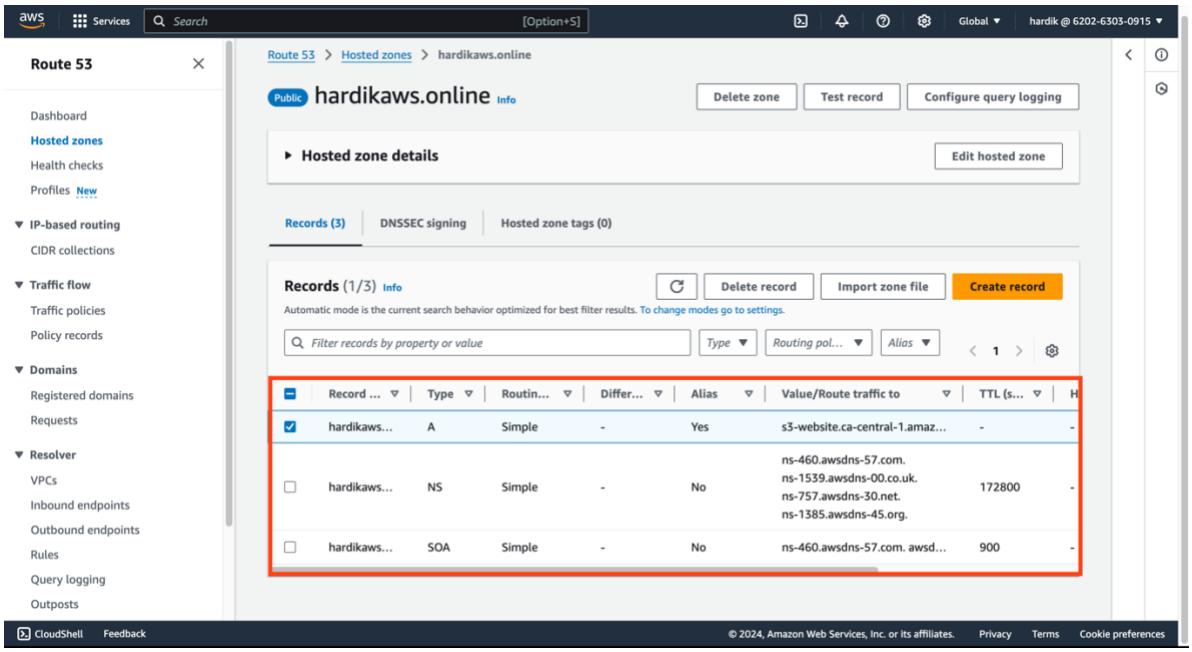


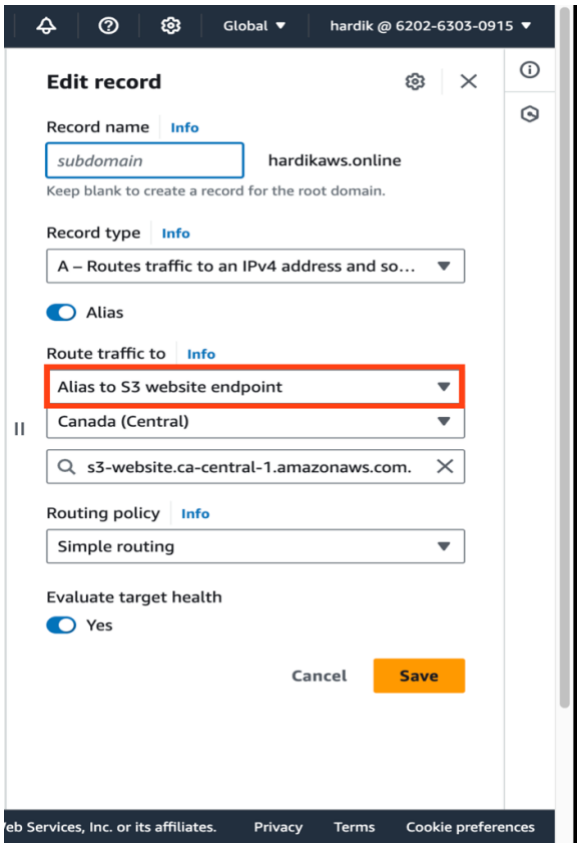
- Established a hosted zone in AWS Route 53 to integrate the third-party domain with AWS.



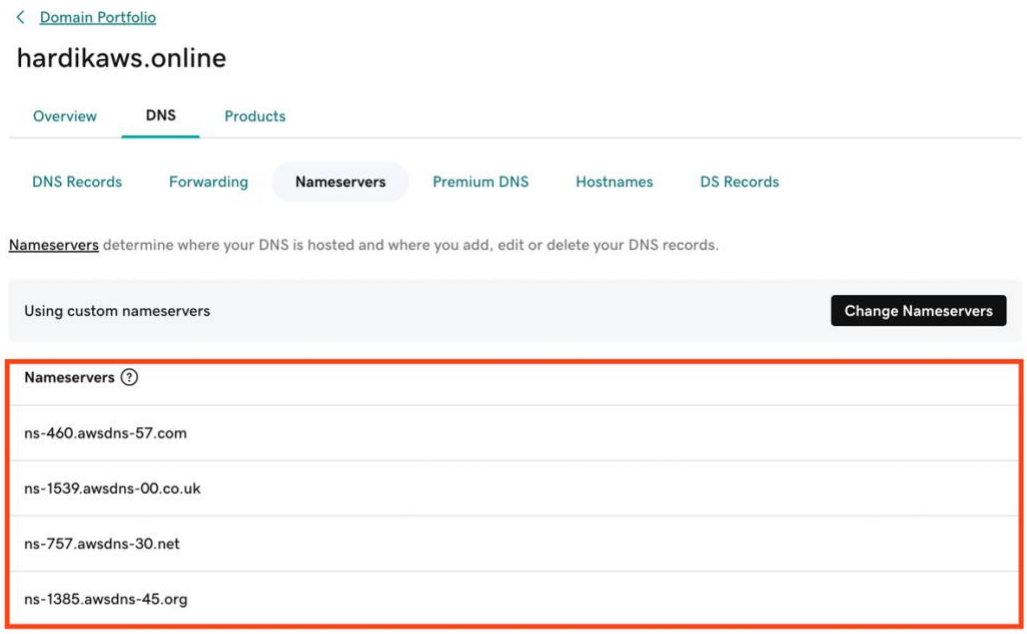
4. DNS Configuration:

- Created DNS records to point to the S3 website endpoint.

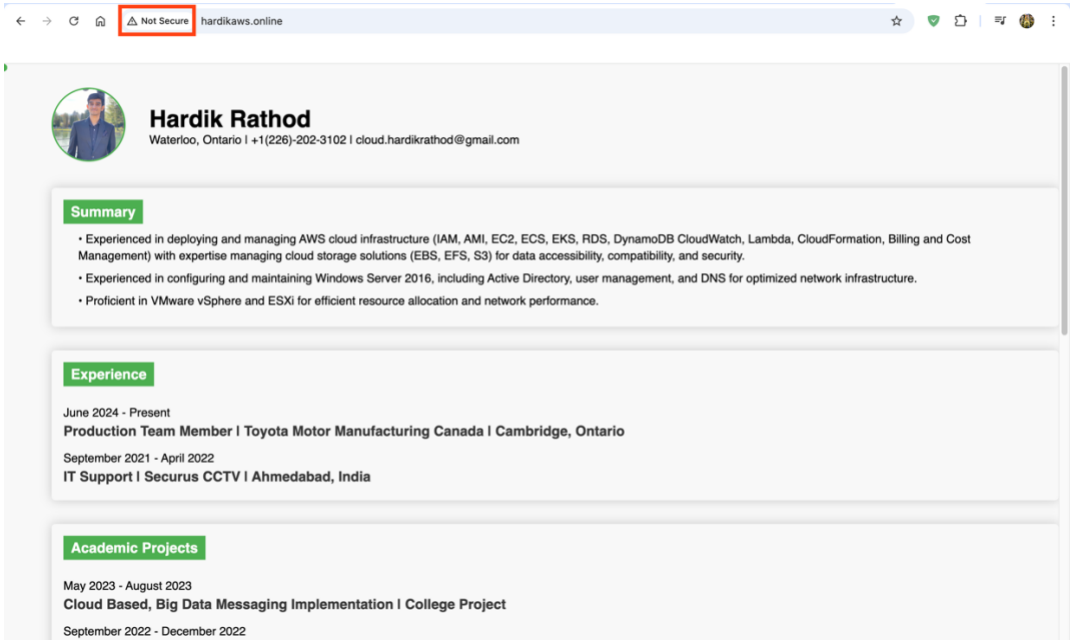




- Updated the name servers in the GoDaddy domain settings to use AWS name servers, ensuring proper domain management.

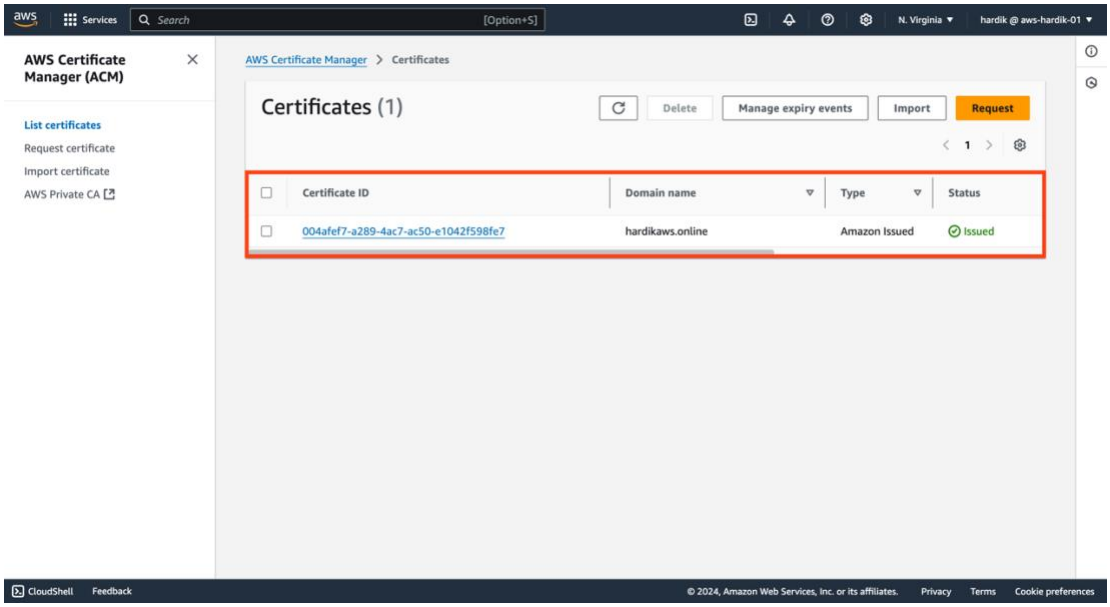


- Accessed the website which is not secured yet.

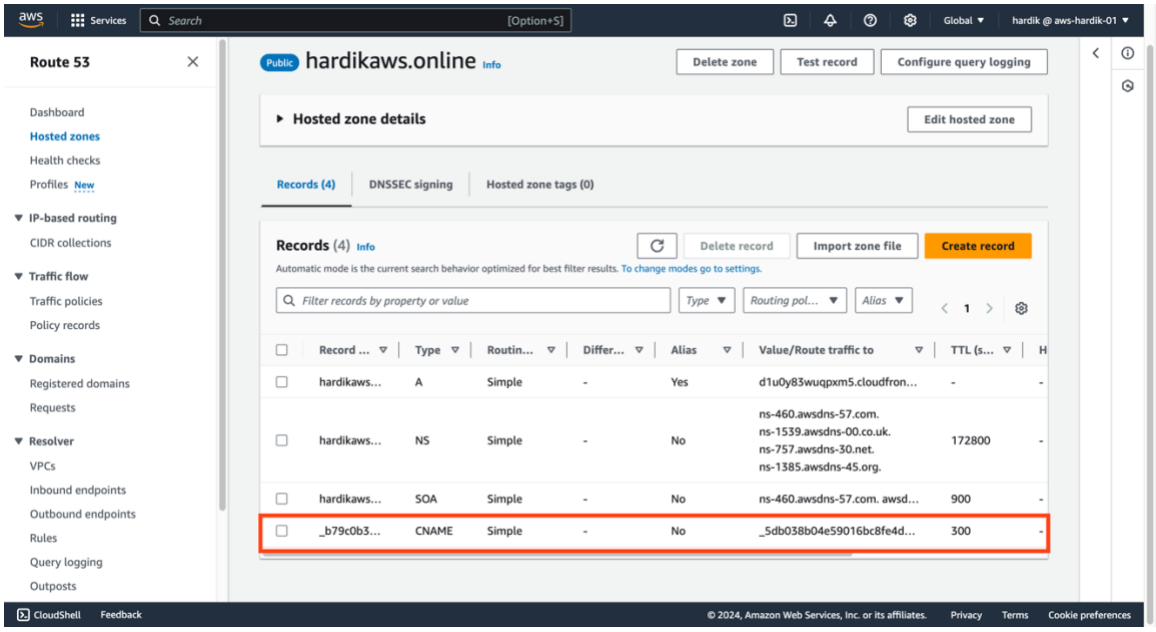


5. SSL Certificate Implementation:

- Obtained an SSL certificate for secure HTTPS connections using AWS Certificate Manager.

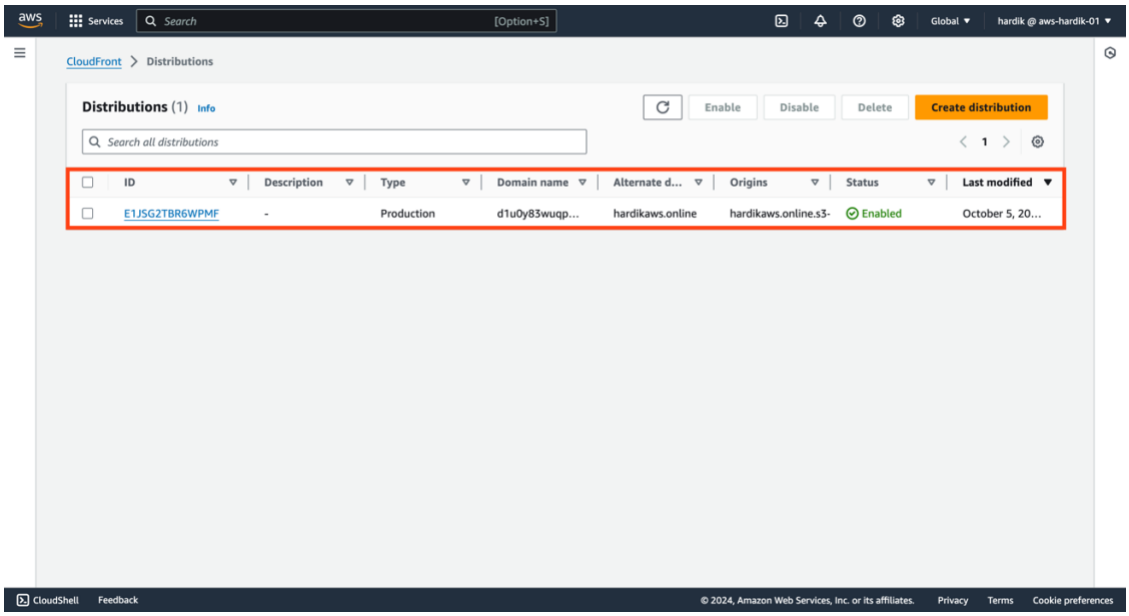


- Validated the SSL certificate using the DNS method for enhanced security.

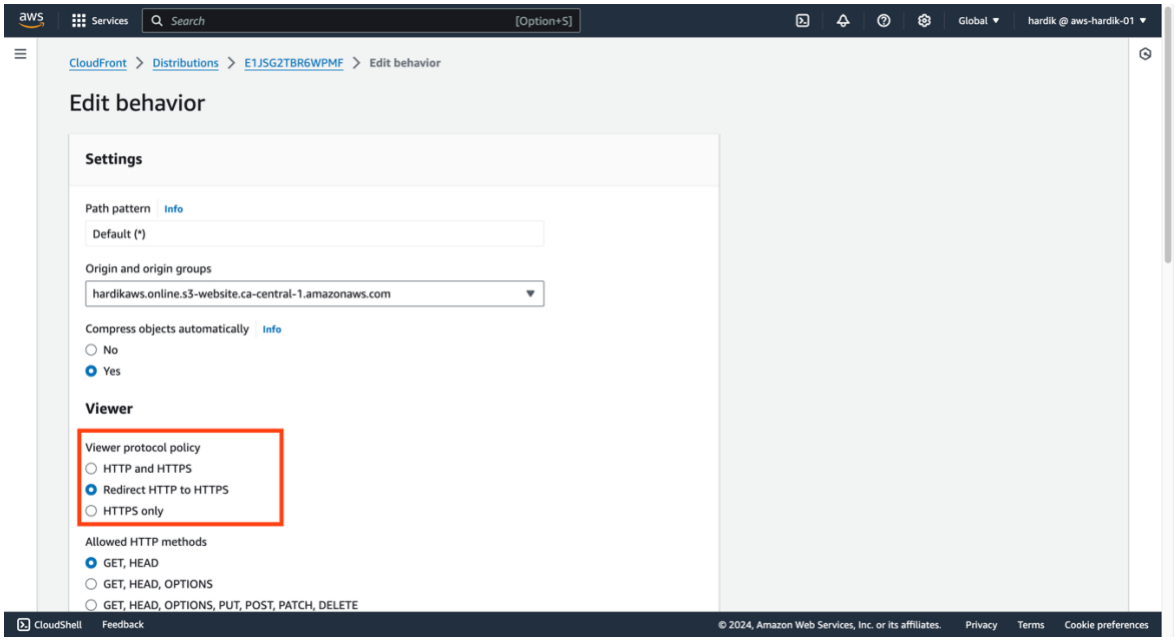


6. CloudFront Distribution:

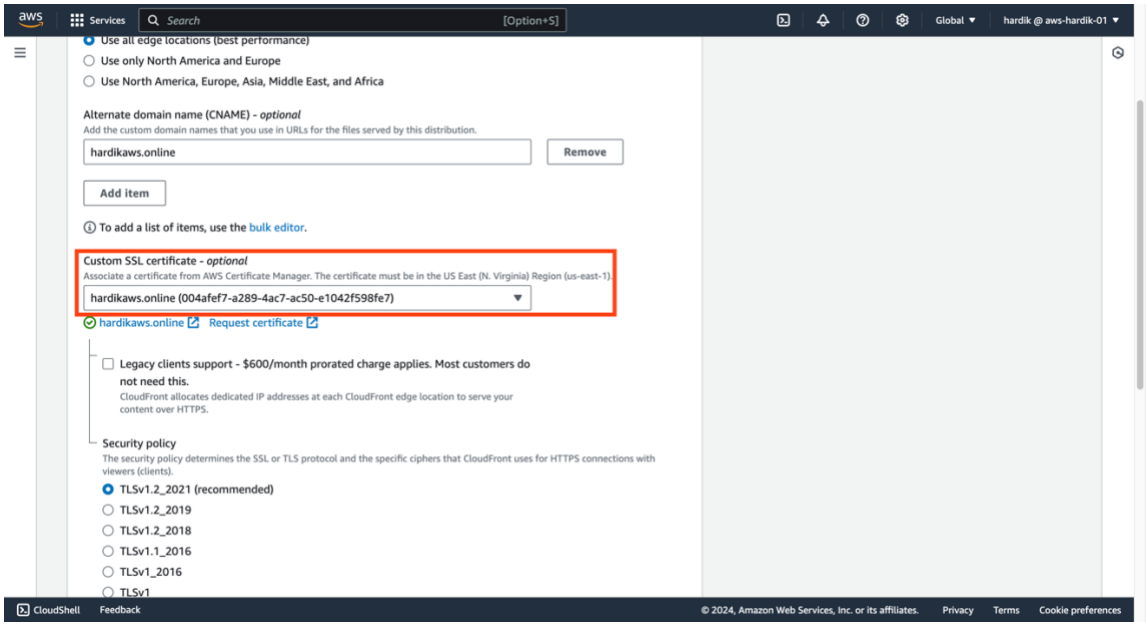
- Created an Amazon CloudFront distribution to serve the S3 website.



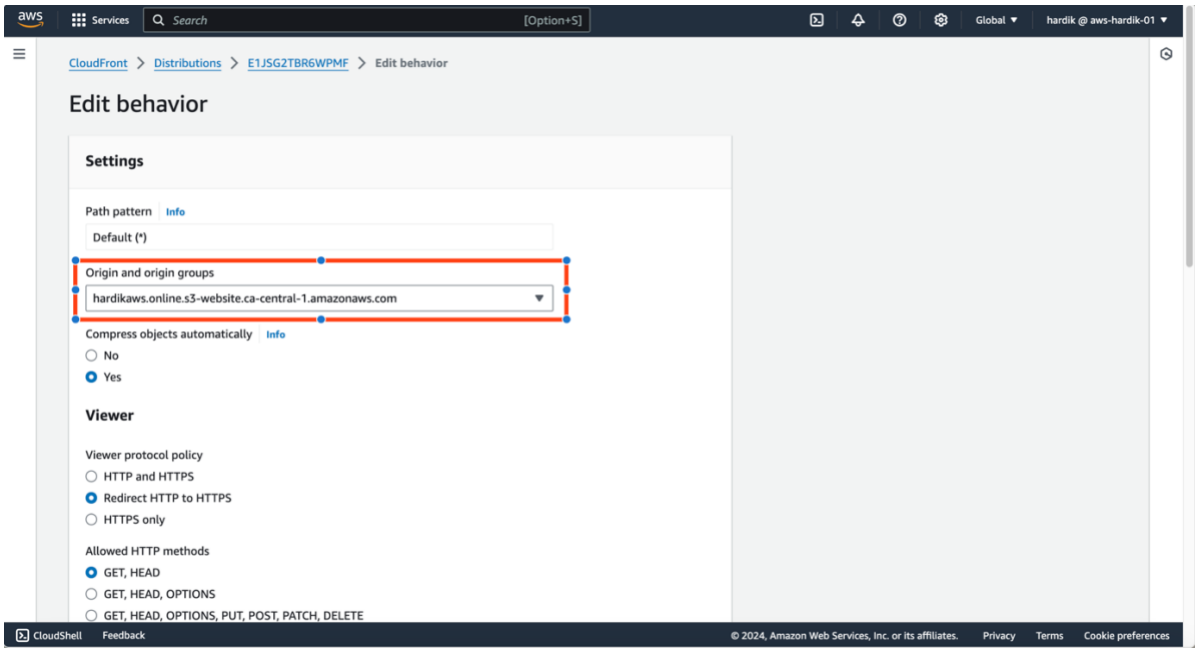
- Configured CloudFront to redirect all HTTP requests to HTTPS, ensuring secure access.



- Attached the SSL certificate to the CloudFront distribution for encrypted data transfer.

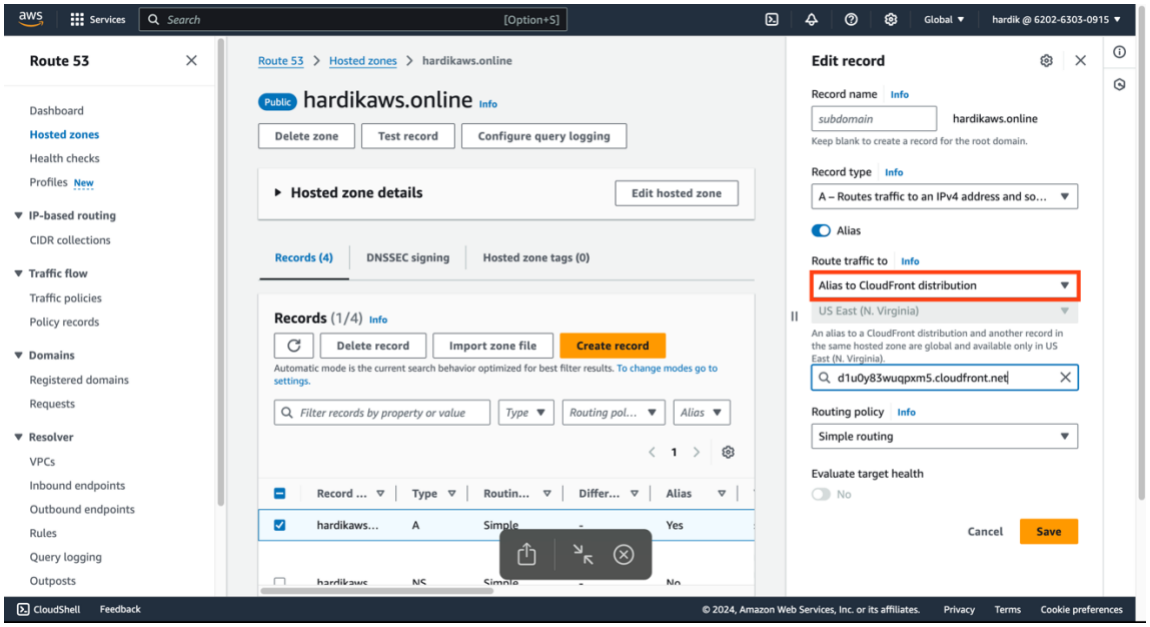


- Set the S3 website endpoint as the origin for the CloudFront distribution.

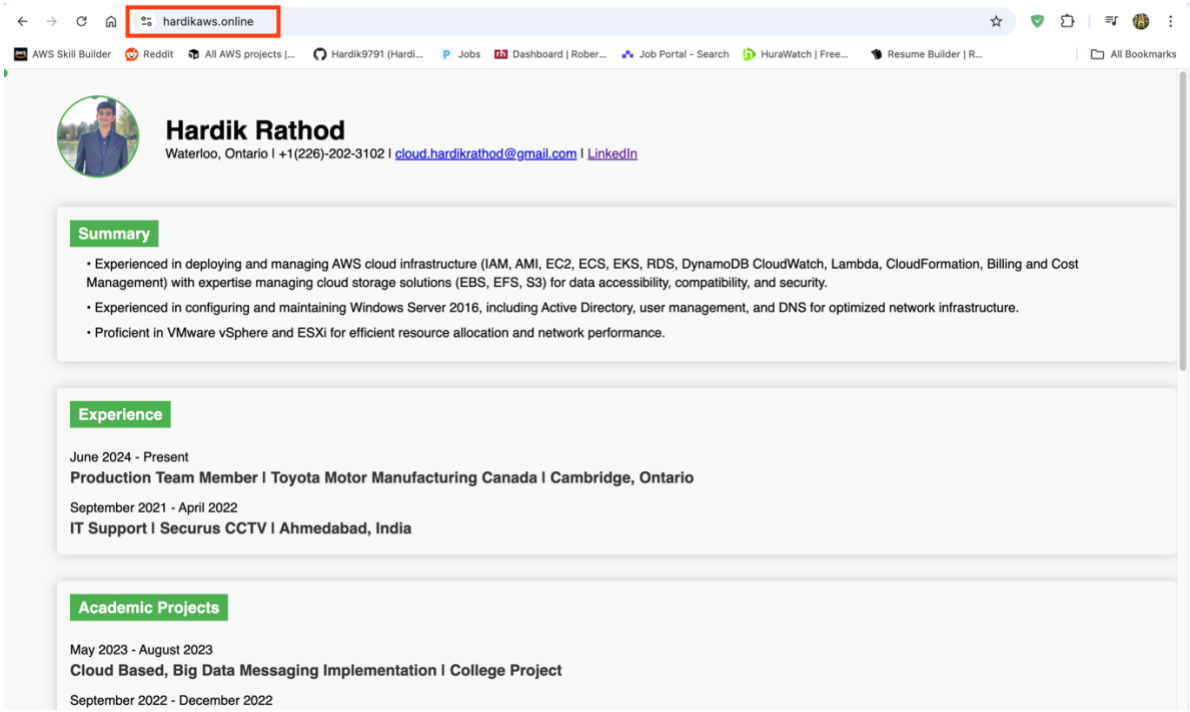


7. Final DNS Configuration:

- Modified the alias configuration in Route 53 to utilize the CloudFront distribution instead of the S3 website endpoint for optimized performance and security.



- Accessed the website which is secured now with the help of CloudFront.



Key Features:

- **Deployment of Web Applications:** Successfully deployed web applications in the cloud environment, ensuring high availability and scalability.
- **AWS Service Integration:** Integrated various AWS services, enhancing application functionality and performance.
- **Security Best Practices:** Implemented security best practices, ensuring data protection and compliance with industry standards.
- **Efficient Resource Management:** Managed cloud resources efficiently, optimizing costs while maintaining performance.

Summary:

This project highlights my ability to deploy and manage web applications in the cloud, showcasing skills in AWS service integration, security best practices, and efficient resource management.